

(1)

$$(a) V_t = R_a I_a + E_a = R_a I_a + K \phi \omega_m$$

@ start, $\omega_m = 0$. If $V_t = \text{rated voltage}$, then I_a is too high as R_a is too small and $\omega_m = 0$.

(b) At start, the field current is to be set @ the rated value $\rightarrow K \phi$ is at the rated/largest value so ω_m won't exceed its rated value after the terminal voltage V_t reaches its rated value.

(c) By adjusting/controlling the motor armature voltage V_t to maintain speed of the motor.

(2)

$$EMF_{rms,ph} = \frac{1}{\sqrt{2}} 2\pi f N \phi = 4.44 f N \phi_{ph}$$

(3)

$$emf = \frac{d\lambda}{dt} \text{ OR } EMF = \omega N \phi = 2\pi f N \phi$$

(a) if speed $\omega + 10\%$, then $V(EMF)$ is $+10\%$ and $f + 10\%$.

(b)

$EMF = \omega N \phi = 2\pi f \phi$, if $\phi + 10\% \rightarrow V + 10\%$ and f no change.

(c) ϕ is proportional to $I_f \rightarrow$ so if $I_f + 10\% \rightarrow V + 10\%$ and f no change

(4)



(a)

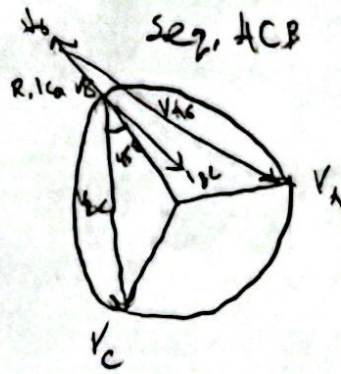
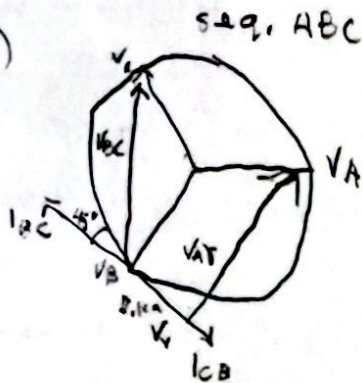
(5) synchronous generator voltage can be regulated by adjusting/controlling the current to the rotor (that is the excitation of the generator)

(6) Synchronous speed is the speed of the rotor is the same as the speed of the stator supply flux (after accounting the # of poles in the synchronous line)

(7) Phase sequence is one phase leads/lags of the other phase, such as ABC (A leads B by 120° , and B leads C by 120°)

To change the phase sequence can be done by exchanging connections of any two phases

(8)



V_{AB} of seq. ABC
 V_{AC} of seq. ACB

(9) Synchronization is the speed of the rotor is the same as the speed of the stator supply flux (after accounting for # of poles in synchronous machine.)

Condition: before synchronization, magnitude + phase sequence + rotating speed/frequency of the 3-phase voltages induced by the rotating rotor flux are the same as those of the 3-phase stator supply voltage

(10) (a) If all 3 lamps go dark/bright at the same time, then the phase sequence is the same as infinite bus.
otherwise $\rightarrow$ if not $\rightarrow$ if lamps go dark/bright in sequence, the phase of the two systems are opposite

(b) All the lamps are dark, other wise it not.

(c) All the lamps go dark at the same time + stay dark for sometime. otherwise it not.

(11) To make the output power factor = 1. Example $\rightarrow$ a normal RL load w/ lagging PF, the PF correction is to inject a leading PF current into the load connection to increase PF to 1.

(12) Inject more real power through the turbine. (in this lab $\rightarrow$ + output power of DC motor).